

RESEARCH ARTICLE

Contrasting temperature effects on the velocity of early- versus late-stage vegetation green-up in the Northern Hemisphere

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Abstract

Global vegetation greening has been widely confirmed in previous studies, yet the changes in the velocity of green-up in each month of green-up period (GUP) remains unclear. Here, we defined the velocity of vegetation green-up as V_{NDVI} (the monthly increase of Normalized Difference Vegetation Index [NDVI] during GUP) and further explored its response to climate change in middle-high-latitude Northern Hemisphere. We found that in early GUP, V_{NDVI} generally showed positive trends from 1982 to 2015, whereas in late GUP, it showed negative trends in most areas. Such contrasting trends were mainly due to a positive temperature effect on V_{NDVI} in early GUP, but this effect turned negative in late GUP. The increase of soil moisture also in part explained the accelerated vegetation green-up, especially in the arid and semi-arid ecosystems of inland areas. Our analyses also indicate that the first month of the GUP was the key stage impacting vegetation greenness in summer. Future warming may continuously speed up the early growth of vegetation, altering the seasonal trajectory of vegetation and its feedbacks to the Earth system.

KEYWORDS

global greening, limiting factor, phenology, spatiotemporal change, terrestrial ecosystem, vegetation dynamic

1 | INTRODUCTION

Global greening, indicated by the satellite-derived vegetation indicators, for example, Normalized Difference Vegetation Index (NDVI) and leaf area index (LAI), is one of the highly credible evidence for anthropogenic climate change (Chen, Park, et al., 2019; Piao et al., 2020; Zhu et al., 2016). This greening phenomenon indicates a major response of terrestrial ecosystems to climate change (Zhu et al., 2016), which also provides multiple feedbacks to Earth systems through biophysical and biogeochemical processes (Cheng et al., 2017; Keenan et al., 2016; Zeng et al., 2017). For example, it has been widely confirmed that greening enhances vegetation photosynthesis (Chen, Ju, et al., 2019; Winkler et al., 2019;

Zhang et al., 2018), as well as terrestrial ecosystem carbon storage (Shevliakova et al., 2013). During the past decades, greening plays an important role in mitigating global warming, offsetting almost one third of anthropogenic CO₂ emission (Friedlingstein et al., 2020). Moreover, greening also significantly impacts hydrological cycle (Lian et al., 2020; Zeng et al., 2018) and alters the land surface energy balance through changes in evapotranspiration, albedo, and other land surface biophysical properties (Arora & Montenegro, 2011; Zeng et al., 2017). Considering large increases in the vegetation greenness during GUP and the widely observed changes in plant phenology, the impacts and feedbacks of the vegetation changes are more predominate during GUP (Buermann et al., 2018; Gonsamo et al., 2017). Therefore, it is important to understand how climate

change regulates the interannual changes of vegetation green-up velocity.

For the temperate ecosystems, vegetation greening (i.e., the interannual increase of the annual integral of NDVI) indicates the enhanced magnitude of green-up ($A_{\text{green-up}}$) during GUP, which can be attributed to the lengthening of GUP ($T_{\text{green-up}}$) and/or the interannual increase of the velocity of leaf green-up ($V_{\text{green-up}}$, interannual increase of $V_{\text{green-up}}$ was defined as green-up acceleration), given that $A_{\text{green-up}} = T_{\text{green-up}} \times V_{\text{green-up}}$ (Wang et al., 2018). On the one hand, climate warming induces the advance of the start of the growing season (SOS; Piao et al., 2006; Zhou et al., 2001), which would lengthen $T_{\text{green-up}}$ if the peak of growing season (POS) remains relatively stable or has a relatively slower advancing speed. On the other hand, climate change may accelerate $V_{\text{green-up}}$ through enhancing activities of enzyme (Kardol et al., 2010) and increasing water and nutrient availabilities (Finger et al., 2016; Keuper et al., 2012; Nemani et al., 2003). Previous studies have generally confirmed the advance of SOS in the past decades (Cleland et al., 2007; Cong et al., 2013; Piao et al., 2019) with an average rate of 2.1 days per decade from 1982 to 2011. Meanwhile, significant advance of POS in northern hemisphere was observed in 19.4% area from 1982 to 2012, with 0.61 days per year in these areas (Xu et al., 2016). These results explored the potential changes in $T_{\text{green-up}}$, but the changes in $V_{\text{green-up}}$ in the past decades still need further investigation.

Wang et al. (2018) found that global average $V_{\text{green-up}}$ increased by $0.003 \text{ m}^2 \text{ m}^{-2} \text{ year}^{-1}$ from 1982 to 2015 (i.e., green-up acceleration). This acceleration of vegetation green-up was observed in most ecosystems except for closed evergreen coniferous forests, lichens and mosses, and sparse vegetation (Wang et al., 2018). However, the overall increase of $V_{\text{green-up}}$ does not fully characterize the dynamic of vegetation because vegetation responses to climate change may be different during each month of the GUP (Zhang et al., 2020; Zhou et al., 2001). The limiting factor of vegetation green-up during different months may vary and changes of vegetation will further feedback to the climate (Lian et al., 2020; Lian et al., 2022), which are important for future climate projection. In addition, the changes of

$V_{\text{green-up}}$ in each month of GUP may show high spatial heterogeneity because of the large differences in biome distribution and their responses to the climate variations (Lian et al., 2020; Piao et al., 2019; Zhang et al., 2020; Zhou et al., 2001). Therefore, exploring the spatiotemporal changes of the velocity of vegetation green-up and identifying the key stage of green-up that mainly influences vegetation productivity are crucial for understanding the climate-vegetation feedbacks and even the biosphere-atmosphere interactions within the Earth system.

Here, we investigated the spatiotemporal change in the velocity of vegetation green-up and its climate drivers in middle-high latitude of the Northern Hemisphere ($>30^\circ\text{N}$). Using NDVI data from Global Inventory Modeling and Mapping Studies (GIMMS), we defined V_{NDVI} (the NDVI increase between two consecutive months during GUP) to indicate the velocity of vegetation green-up at monthly scale (Figure 1). This method was firstly developed by Piao et al. (2022) and successfully used in temperate China to explore the speed of canopy development and senescence. Consequently, it allowed us to test our first hypothesis: The inter-annual trends in V_{NDVI} would vary among different months of GUP. Via performing partial correlation analysis between V_{NDVI} and climate factors (temperature, soil moisture and radiation), we further tested the second hypothesis: The different trends in V_{NDVI} in each month of GUP could result from their different responses to climate change. Furthermore, we classified the study areas according to the interannual changes of V_{NDVI} at different months of GUP to explore the key stage that mainly affects the trends in annual peak NDVI.

2 | MATERIALS AND METHODS

2.1 | Study area

This study focused on the middle-high latitude of the Northern Hemisphere ($>30^\circ\text{N}$) where vegetations have strong seasonal changes. Croplands and evergreen forests were excluded based on

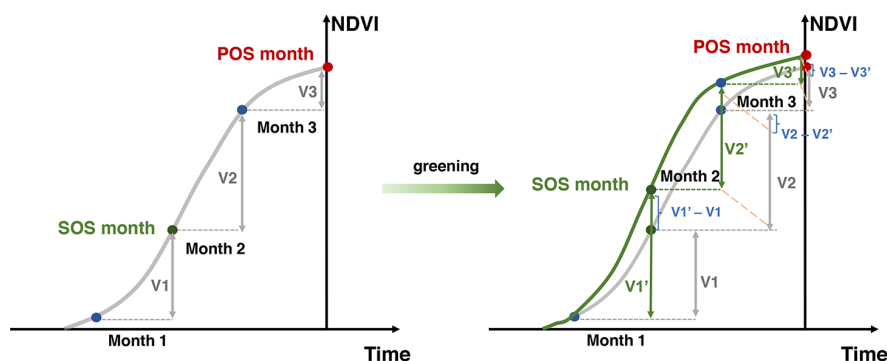


FIGURE 1 Conceptual diagram of velocity of vegetation green-up (V_{NDVI}) and its interannual change. Green-up period in this study is defined as the period from start of the growing season (SOS) month to peak of growing season (POS) month. V_{NDVI} is calculated as the monthly NDVI increase during this green-up period, with V1 indicating the velocity for the SOS month. Three V_{NDVI} values are extracted in this example. Vegetation greening (right panel) indicates the interannual increase of NDVI, but V_{NDVI} (marked as V1', V2', and V3' after greening) in different months exhibit divergent changes. The diagram shows the scenario that V1 increases (green-up acceleration) while V2 and V3 decreases (green-up deceleration). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

the land cover classification of MCD12C1 v006 (Friedl et al., 2010). Moreover, area covered with bare soil/sparse vegetation (annual mean NDVI below 0.1) was also excluded from our analysis (Zhou et al., 2001).

2.2 | Datasets

NDVI data set of GIMMS from 1982 to 2015 (Tucker et al., 2004, 2005) was used to indicate vegetation greenness in this study, which has been refined and corrected for orbital drift, calibration, viewing geometry, and volcanic aerosols (Kaufmann et al., 2000). The spatial and temporal resolutions of this dataset were 0.083° and biweekly, respectively. The maximum value composite on each month's biweekly records was used as the monthly NDVI data to reduce the residual atmospheric effect. Moderate-resolution Imaging Spectroradiometer (MODIS) enhanced vegetation index (EVI) product MOD13C2 (Didan, 2015) from 2000 to 2015 was also used to validate the results based on GIMMS NDVI.

The mean 2-m surface temperature was acquired from the CRU. TS4.05, with a spatial resolution of 0.5° and a monthly temporal resolution (Harris et al., 2020). Data of solar radiation was acquired from CRU-JRA v2.2, a combination of CRU and Japanese reanalysis dataset (JRA), with a 0.5° spatial and 6-hour temporal resolution (<https://catalogue.ceda.ac.uk/uuid/7f785c0e80aa4df2b39d068ce7351bbb>; Harris et al., 2014; Kobayashi et al., 2015). Moreover, surface soil moisture was acquired from the C3S dataset provided by European Center for Medium-Range Weather Forecasts (ECMWF), with a 0.25° spatial and monthly temporal resolution (<https://cds.climate.copernicus.eu/cdsapp#!/dataset/satellite-soil-moisture?tab=overview>).

2.3 | Data analysis

In this study, GUP was defined as the period from the month with SOS to the month with POS. We employed the double logistic (Dlog) method to derive SOS for each pixel. This method used a double logistic function to fit the original NDVI time series for each year and the SOS can be directly retrieved from the coefficients the fitted curve (Meng et al., 2021; Zhang et al., 2020). POS is defined as the month with maximum NDVI. For a given pixel, we obtained the SOS and POS month of each year and then used the mode (the number appearing for the most times) during 1982–2015 as the SOS/POS month of this pixel. The length of GUP was the duration from SOS month to POS month (Figure S1). For example, for a pixel with SOS in March and POS in May, we calculated V_{NDVI} for March as NDVI in March minus NDVI in February for each year (Figure 1). V_{NDVI} for April and May can be calculated in a similar way. We marked March, April and May as the first, second and third months in GUP, respectively for this pixel.

During the analysis, we firstly excluded the negative NDVI data and then calculated V_{NDVI} . For the month before SOS, we used multiyear mean NDVI in place of the actual NDVI to exclude potential

snow effect. For each pixel, we conducted linear regression to explore the interannual trend in V_{NDVI} and NDVI. The relative trend in V_{NDVI} was calculated using trend in V_{NDVI} divided by multiyear averaged V_{NDVI} . To further explore the climate drivers (temperature, soil moisture and radiation) of the trends in V_{NDVI} and NDVI, we performed partial correlation analysis to remove the covariate effects among input variables, which has been widely applied in previous studies involving climate change and vegetation phenology (Fu et al., 2015; Peng et al., 2013). Considering the potential lagged effects of climate variables on V_{NDVI} and NDVI (Fu et al., 2015; Menzel et al., 2006), we used pre-season climate rather than the monthly climate corresponding to V_{NDVI} /NDVI in the partial correlation analysis (see Figures S2 and S3 for the length of pre-season). The pre-season was defined as the period with the highest Pearson correlation coefficient between climatic factors and V_{NDVI} (or NDVI; Gao et al., 2019). Pre-season was calculated for up to 3 months prior in this study (Piao et al., 2006; Rundquist & Harrington, 2000). It is noteworthy that we resampled the climate data to 0.083° to match the resolution of NDVI during analyses. We also repeated above data analyses using MODIS EVI data to validate the results based on GIMMS NDVI (Figures S4–S6).

Furthermore, we classified the study areas into seven groups based on different combination of monthly V_{NDVI} trends. During the process of classifying, we first identified the times (T_0) that V_{NDVI} showed opposite trends between two consecutive months. If T_0 is 0, it indicates that all the months showed monotonical trends in this pixel, so that we classified this pixel into group of “+” (“–”). If T_0 is 1, we classified this pixel into group of “+ –” (“– +”), which indicated V_{NDVI} showed positive interannual trend in early GUP but negative trend in late GUP (and vice versa). Similarly, we classified group of “+ – +” and “– + –” when T_0 is 2. The pixels with T_0 larger than 2 were classified into “other” group.

3 | RESULTS

3.1 | Spatiotemporal changes in the velocity of vegetation green-up

An overall greening trend was observed in the study region from 1980s to 2010s (Figure 2a). Specifically, annual maximum NDVI ($NDVI_{max}$) showed positive trends in 67.41% areas, with 39.09% being significant ($p < .05$, Figure 2b). The increase in $NDVI_{max}$ could result from the enhanced V_{NDVI} in different months of GUP, which was further explored below.

As shown in Figure 3a, areas with longer GUP usually presented smaller V_{NDVI} , with smaller variation of V_{NDVI} among different months, indicating a slow green-up process in these areas, for example, Europe and USA (Figure 4a–f). The highest velocity of green-up generally occurred in the second month of GUP (Figure 3a). In contrast, for the regions with shorter GUP (e.g., Siberia and the northernmost area of North America), monthly V_{NDVI} was much larger and showed greater variation

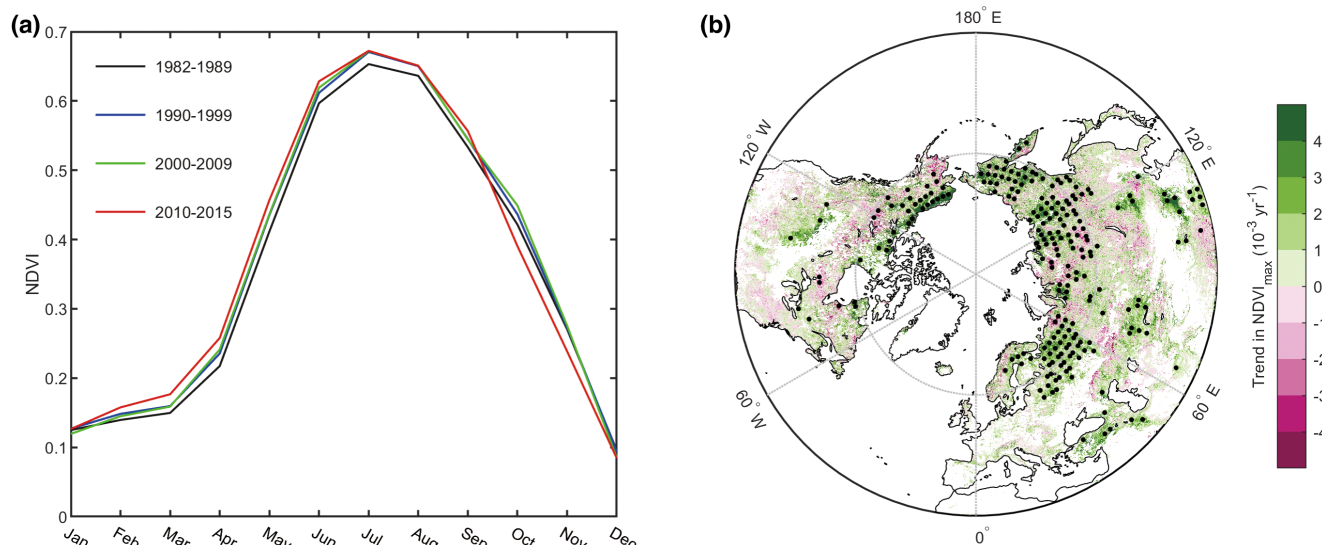


FIGURE 2 Changes in NDVI from 1982 to 2015. (a) Seasonal change in monthly NDVI. The NDVI curves are derived from the mean values of all pixels across the study region. (b) Spatial pattern of the trends in annual maximum NDVI. The regions labeled with black dots represent locations with a significant trend in NDVI ($p < .05$). [Colour figure can be viewed at wileyonlinelibrary.com]

among different months (Figures 3a and 4a–f), suggesting a rapid green-up. The largest V_{NDVI} was generally observed in the first month of GUP for these areas.

We further explored the interannual change in monthly V_{NDVI} and found divergent trends across different months of GUP (Figures 3b,c and 4g–r). For example, positive trends in V_{NDVI} were generally observed in Europe during March (Figure 4g). The positive trends spread to higher latitudes and inland regions in April (Figure 4h), with the onset of spring phenology in more areas (Figure S1a). Meanwhile, significantly positive trends in V_{NDVI} were also observed in the central North America (Figure 4h). During April, acceleration of vegetation green-up (positive interannual trend in V_{NDVI}) reached the peak and occurred in 71.46% areas. Due to the late SOS, positive trends in V_{NDVI} was observed in Siberia in May (Figure 4i) and in the northernmost region in June (Figure 4j). These results suggest that the velocity of green-up generally accelerated in early GUP. The overall results of trends in V_{NDVI} confirms this phenomenon (Figure 3b): The largest positive trends in V_{NDVI} were always observed in the first month regardless of the length of GUP. However, the trends in V_{NDVI} turned to negative in May for Europe and USA (Figure 4i), where spring phenology started relatively earlier. The negative trends in V_{NDVI} persisted to June and July in Eurasia (Figure 4j,k). These results suggested a general deceleration (negative interannual trend in V_{NDVI}) in late GUP. The robustness of above results was further confirmed by analyses using MODIS EVI (Figures S4 and S5).

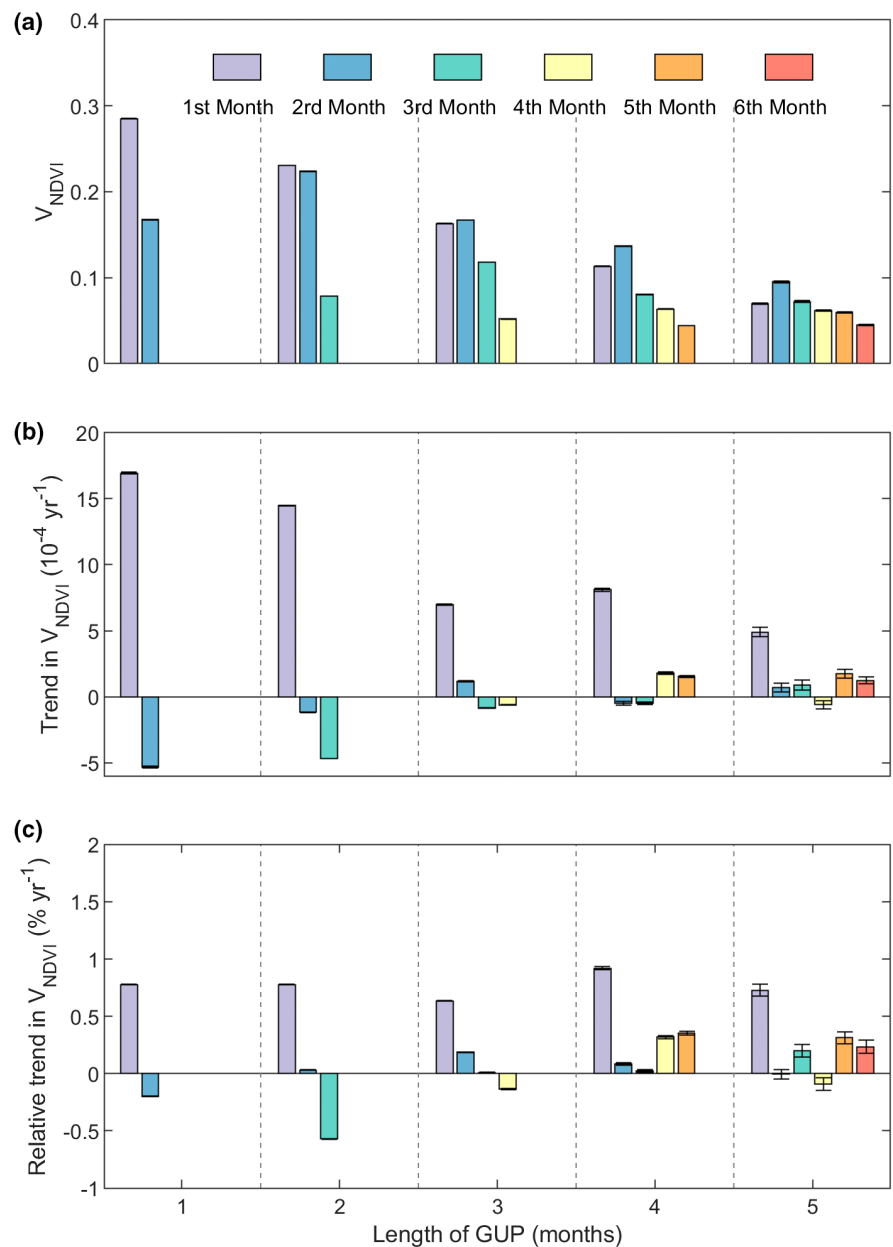
The divergent trends in V_{NDVI} regulate the trends in monthly NDVI (Figure S7). Due to the acceleration of green-up in early GUP, we observed significant greening trends (positive interannual trends in monthly NDVI) during March and April in Europe (Figures S7a,b). However, the greening trends became weaker in May (Figure S7c),

which resulted from the significantly negative trends in V_{NDVI} during May. In other words, although green-up decelerated in late GUP, the enhanced velocity of green-up in early GUP still resulted in an overall greening trend in Europe. In contrast, we did not observe widespread greening trends in North America during March to May, and some areas even exhibited significantly negative trends in monthly NDVI, which was due to the weak positive (or even negative) trends in V_{NDVI} .

This divergent contributions of V_{NDVI} in different months to vegetation greenness were further confirmed as shown by Figure S8. We showed the month with maximum V_{NDVI} (M_{mv}) in Figure S8a and the month with the maximum trend in V_{NDVI} (M_{mtv}) in Figure S8b. M_{mv} showed a clear latitudinal pattern, changing from April in low latitude to June in high latitude (Figure S8a). M_{mv} usually occurred in the SOS month (M_{SOS}) in areas with short GUP but the month following M_{SOS} in areas with long GUP (Figure S8d), which again suggested different green-up paces. M_{mtv} also showed latitudinal pattern (Figure S8b), which occurred at SOS month for 46.25% areas (Figure S8e). Comparing with M_{mv} , M_{mtv} occurred earlier in 22.33% areas, simultaneously in 43.97% areas, and later in 33.70% areas (Figure S8g).

Furthermore, given the different V_{NDVI} in each month of GUP, we also explored the relative trends ((interannual trend in $V_{\text{NDVI}})/(\text{multiyear average } V_{\text{NDVI}}))$ in V_{NDVI} (Figures 3c and 4m–r). Interestingly, although the trend in V_{NDVI} during the late GUP was much smaller than that in early GUP (Figure 3b), the relative trends of them were comparable (Figure 3c). Consequently, the month with maximum relative trend in V_{NDVI} (M_{mrt}) was not always same to M_{mtv} , which was later in 35.74% areas (earlier in 5.01% areas, simultaneously in 59.25% areas, Figure S8i). It indicates although V_{NDVI} in late GUP is relatively smaller than that in early stage, it is also sensitive to climate change.

FIGURE 3 V_{NDVI} in each month during the green-up period (GUP) and its interannual dynamics. (a) The multiyear averaged V_{NDVI} . (b) The interannual trends in V_{NDVI} . (c) The relative trends in V_{NDVI} . The relative trend was defined as percentage of trend in V_{NDVI} to the multiyear averaged V_{NDVI} . Data were grouped based on the length of GUP. Median value of each group was shown as the bars. Error bars indicate the standard errors of the spatial variations. [Colour figure can be viewed at wileyonlinelibrary.com]



3.2 | Climate drivers of the interannual changes in the velocity of vegetation green-up

We further performed partial correlation analysis to explore how climate factors (temperature, soil moisture and radiation) drove the changes in the velocity of vegetation green-up. Figure 5 shows that temperature played a key role in regulating the interannual trends in V_{NDVI} , but it presented opposite impacts between the early and late GUP. In March, positive correlation between temperature and V_{NDVI} was observed in 94.02% area, with 58.07% being significant (Figure 5a). This significant and positive effect of temperature on V_{NDVI} spread to higher latitudes and inland regions in April with the onset of spring phenology in more areas (Figure 5b, positive correlation in 85.41% area, with 53.39% being significant). However, much of the positive effect reversed into negative in these areas during May (Figure 5c). At the same time, soil moisture showed positive effect on V_{NDVI} , especially

for the inland of Eurasia (Figure 5i), where vegetation growth was more constrained by water availability. In the high-latitude areas like Siberia, where SOS occurred in May, temperature still showed positive effect on V_{NDVI} (Figure 5c). In June, the negative effects of temperature were observed in most area (Figure 5d, negative correlation in 64.78% area, with 24.93% being significant). The same results were observed in July (Figure 5e), during which temperature presented negative effect on V_{NDVI} in 78.08% areas, with 28.07% being significant. Such divergent effects of temperature on V_{NDVI} were clearer as presented in Figure 5s, where positive effects were observed in the first month of GUP. In contrast, for the following months, negative effects of temperature on V_{NDVI} were generally observed. It is noteworthy that soil moisture still positively affected V_{NDVI} in the inland of Eurasia and North America during May and June (Figure 5i,j), suggesting the essential role of water supply in regulating vegetation growth in arid and semiarid ecosystems. In comparison with temperature, radiation showed relatively

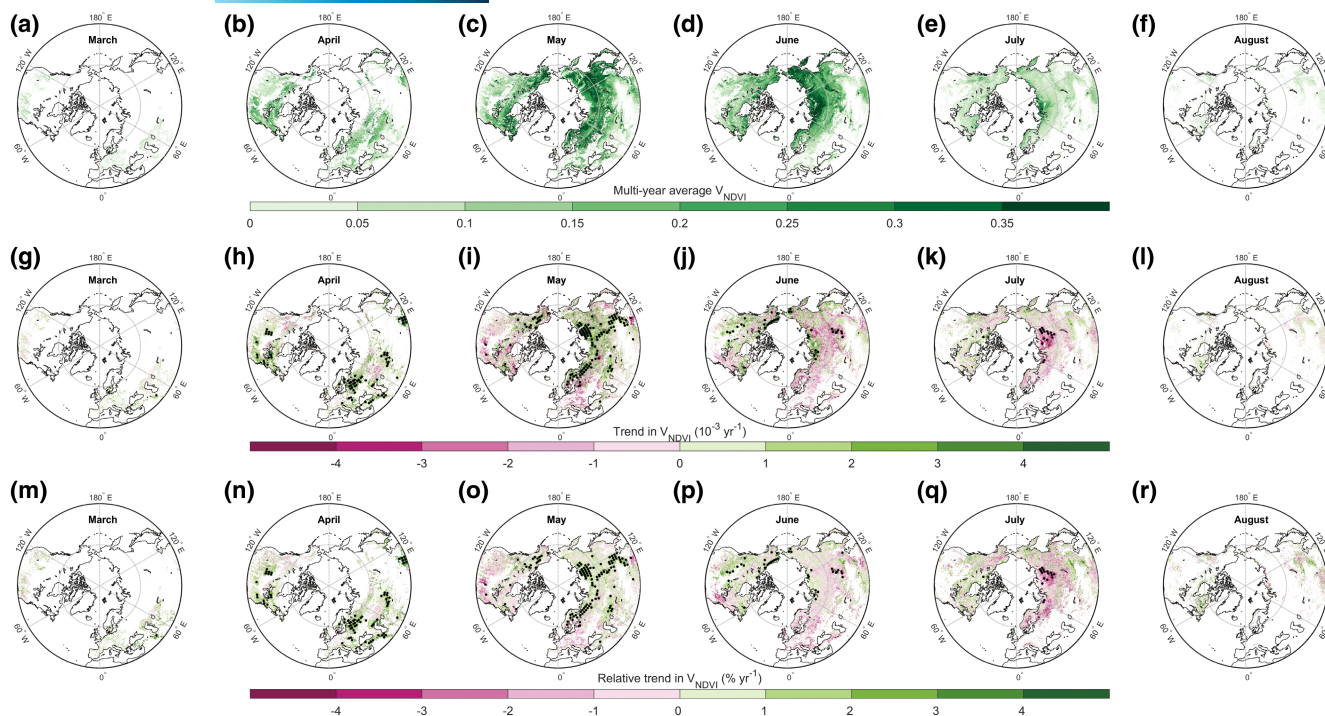


FIGURE 4 Spatial patterns of V_{NDVI} and its trends from March to August. (a–f) Spatial patterns of multiyear average V_{NDVI} . (g–l) Spatial patterns of the trends in V_{NDVI} . (m–r) Spatial patterns of the relative trends in V_{NDVI} . The relative trend was defined as percentage of trend in V_{NDVI} to the multiyear averaged V_{NDVI} . The regions labeled with black dots represent locations with a significant trend in V_{NDVI} ($p < .05$). [Colour figure can be viewed at wileyonlinelibrary.com]

weaker impacts on V_{NDVI} over the entire GUP, and no clear spatial pattern of its impact was observed (Figure 5m–r). Analyses based on EVI also supports the above findings (Figure S6).

Consistent with V_{NDVI} , the temporal dynamics of monthly NDVI were also mainly regulated by temperature (Figure S9), but the positive effects of temperature on NDVI were stronger and lasted for a longer period of time than on V_{NDVI} (Figure 5t). The positive correlation between temperature and NDVI were observed in 95.33%, 94.95%, 91.77% and 86.06% areas in March, April, May and June, respectively (Figure S9a–d). Negative effects of temperature on NDVI were mainly observed in arid and semiarid areas in the inland of Eurasia in June and Siberia in July (Figure S9d,e). Soil moisture also showed positive effect on monthly NDVI in the inland of Eurasia, which were stronger than that on V_{NDVI} (Figure S9i–l). Similar to V_{NDVI} , the impacts of radiation on NDVI were weak and showed high spatial heterogeneity (Figure S9m–r).

3.3 | Portfolio of changes in the velocity of vegetation green-up

To further dissect the changes in green-up paces, we classified the study areas into seven groups based on the interannual trends in V_{NDVI} in different months of GUP (Figure 6). Type “+ –” indicates vegetation green-up accelerated in early GUP but decelerated in late GUP, which accounted for 45% areas (Figure 6b). This type mainly distributed in Europe, Siberia, and northwest of North America (Figure 6a),

where significant increase of peak NDVI was observed (Figure 2a). Another two groups including “+” (acceleration in all months) and “+ – +” (acceleration in early GUP, then deceleration, finally acceleration) accounted for 11% and 15%, respectively (Figure 6b). They mainly distributed in central Eurasia and central North America (Figure 6a), where peak NDVI also significantly increased in past decades (Figure 2a). In contrast, the types indicating deceleration of green-up in early GUP including “–” (deceleration in all months), “– +” (deceleration in early GUP but acceleration in late GUP), and “– + –” (deceleration in early GUP, then acceleration, finally deceleration in late GUP) generally distributed in areas where we observed negative trend in peak NDVI (Figures 2a and 6a). Typically, type “– +” mainly distributed in Canada, which accounted for 13% of study areas (Figure 6b). The remaining three groups including “–”, “– + –” and “other” accounted for 3%, 9% and 4% areas, respectively (Figure 6b).

We further compared the dynamics of $NDVI_{max}$ between group with increasing V_{NDVI} in early GUP (“+”, “+ –” and “+ – +”) and group with decreasing V_{NDVI} in early GUP (“–”, “– +” and “– + –”) (Figure 6c). For the group with increasing V_{NDVI} in early GUP, we observed positive trend in $NDVI_{max}$ in 72.67% (44.41% being significant) areas, with an average trend of 0.0010 year^{-1} . In contrast, for the group with decreasing V_{NDVI} in early GUP, we only observed positive trend in $NDVI_{max}$ in 53.89% (26.09% being significant) areas. The average trend in $NDVI_{max}$ in this group was 0.0002 year^{-1} , significantly smaller than that of group with increasing V_{NDVI} in early GUP ($p < .001$, one-way analysis of variance). Comparison between type “+ –” and “– +” also confirmed such difference (Figure 6c).

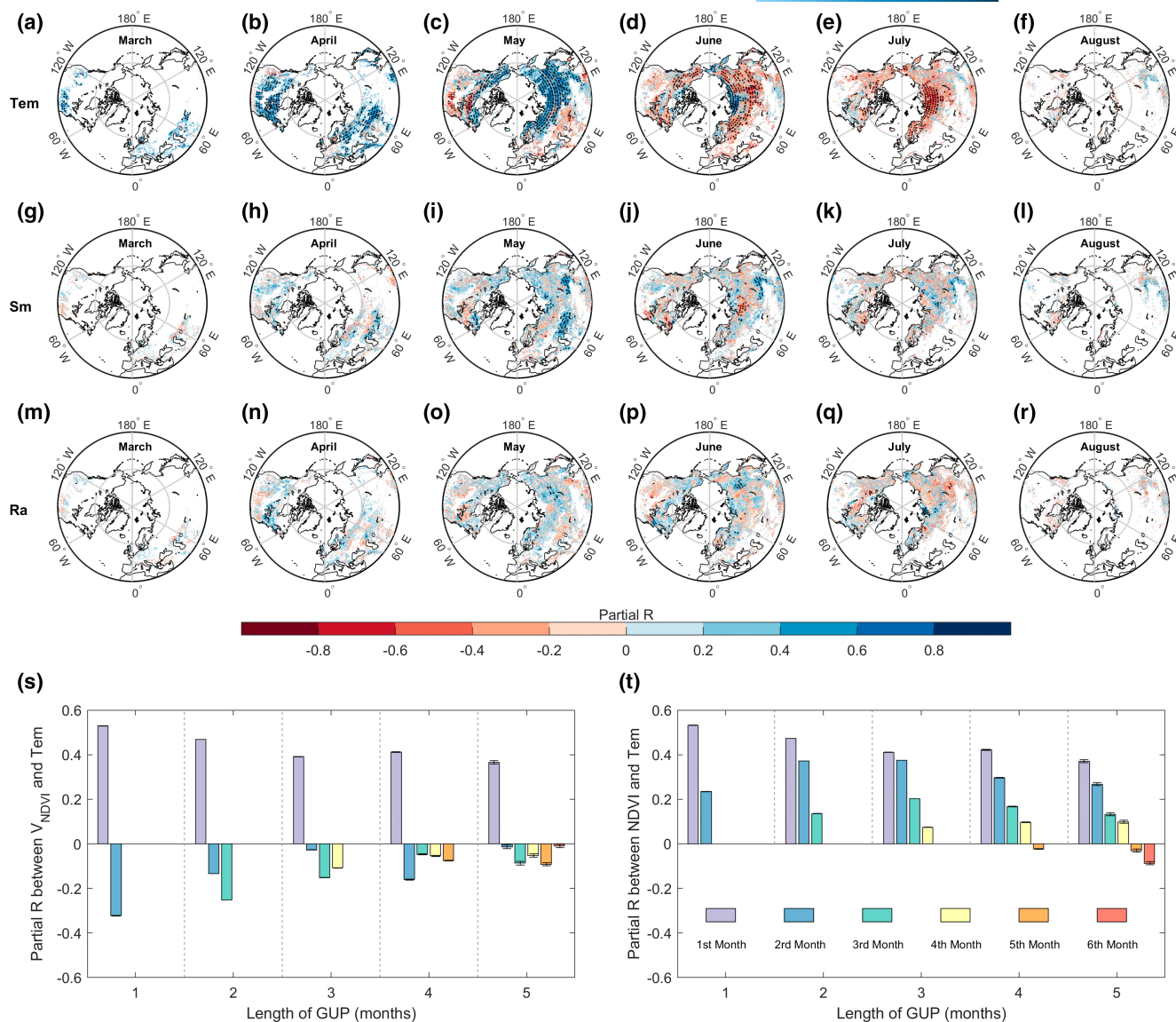


FIGURE 5 The effects of climate change on V_{NDVI} and NDVI. (a–f) The spatial patterns of the partial correlation coefficients between V_{NDVI} and temperature (Tem) after controlling for soil moisture (Sm) and radiation (Ra). (g–l) The spatial patterns of the partial correlation coefficients between V_{NDVI} and Sm after controlling for Tem and Ra. (m–r) The spatial patterns of the partial correlation coefficients between V_{NDVI} and Ra after controlling for Tem and Sm. The regions labeled with black dots represent locations with significant partial correlation coefficients ($p < .05$). (s) Partial correlation coefficients between V_{NDVI} and temperature (Tem) after controlling for soil moisture and radiation. Corresponding preseason for the climate factors used in (a–s) were shown in Figure S2. (t) Partial correlation coefficients between NDVI and temperature (Tem) after controlling for soil moisture and radiation. Mean value of each group was shown as the bars. Panel s shares color bars with (t). Error bars indicate the standard errors. Corresponding preseason for the climate factors used in this panel were shown in Figure S3. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Such patterns become more evident if we converted the calendar month to phenological month (Figure S10). The first month of GUP (i.e., SOS month) showed the most substantial interannual trend (Figure S10a), which exhibited similar spatial pattern to the trend in $NDVI_{max}$ (Figure 2a). Positive trend in V_{NDVI} for the SOS month was observed in 72.97% areas, with 52.75% having increasing $NDVI_{max}$. That is, for areas showing acceleration of vegetation green-up in the SOS month, 72.29% of them showed increasing $NDVI_{max}$. The above results indicate the acceleration of vegetation green-up in early GUP, especially the first month, is the major driver of increasing NDVI in summer.

4 | DISCUSSION

Many previous studies have explored the changes in NDVI induced by climate change and land use management (Chen, Park, et al., 2019; Piao et al., 2020; Zhu et al., 2016), but studies focusing on the spatiotemporal changes in V_{NDVI} are rather limited (Wang et al., 2018). In this study, we observed an inconsistency between the changes in NDVI and V_{NDVI} : NDVI generally showed positive inter-annual trends in most months during GUP, while the trends in V_{NDVI} varied across different months. Specifically, vegetation green-up generally

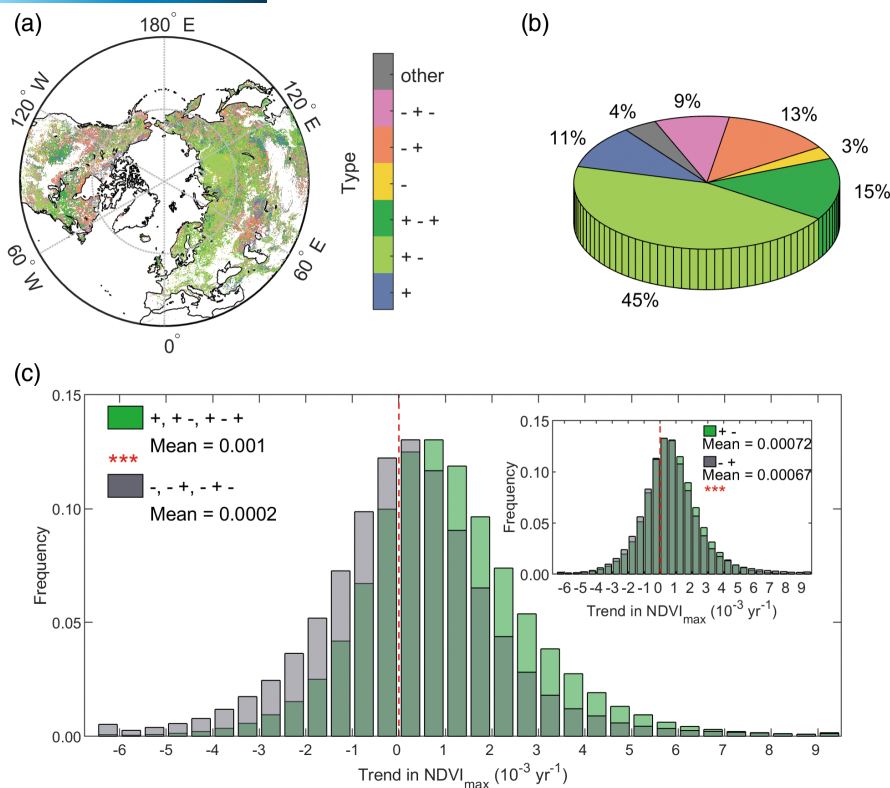


FIGURE 6 Different types of V_{NDVI} dynamics. (a) The spatial distribution of each type. (b) The percentage of pixels for each type. Type “+” indicates V_{NDVI} accelerated (positive interannual trend) in all the months during the green-up period. Type “+ -” indicates V_{NDVI} accelerated in the early green-up period and then decelerated in late green-up period. Type “+ - +” indicates V_{NDVI} accelerated in the early green-up period and then decelerated, but again accelerated in late green-up period. The remaining types were defined similarly (see Section 2). (c) Frequency distributions of the trend in annual maximum NDVI ($NDVI_{max}$) in different groups. Green bars indicate the group combined from type “+”, “+ -” and “+ - +”, whereas gray bars indicate the group combined from type “-”, “- +” and “- + -”. “***” indicates the significant difference ($p < .001$) between two groups based on one-way analysis of variance (ANOVA). The inset shows the comparison between type “+ -” and “- +”. [Colour figure can be viewed at wileyonlinelibrary.com]

accelerated in early GUP but decelerated in late GUP, which confirms our first hypothesis. The different inter-annual trends between NDVI and V_{NDVI} is mainly due to that NDVI represents the state of canopy, but V_{NDVI} represents the process of canopy development, which is directly related to the photosynthetic carbon assimilation and partitioning. Therefore, V_{NDVI} could be more sensitive to the climate change at monthly scale and can be a better indicator of sub-seasonal vegetation dynamics than NDVI. NDVI, on the other hand, depends not only on vegetation responses to recent climate (V_{NDVI}), but also its status of the previous timestep. Further analyses confirmed above views. Although both NDVI and V_{NDVI} were dominantly regulated by temperature, comparing with the overall positive response of NDVI to increasing temperature, V_{NDVI} showed divergent responses to climate warming in early versus late GUP. In early GUP, climate warming generally accelerated vegetation green-up, but in late GUP, warming decelerated green-up paces, which confirms our second hypothesis. The observed declining temperature dependency of NDVI may be explained by a strong positive temperature effect on V_{NDVI} during the early GUP, whose carryover effect is progressively enervated by the negative temperature effect on V_{NDVI} for later GUP.

The divergent responses of V_{NDVI} to climate warming may be due to different resource limitations of vegetation growth in early and late GUP. During the early GUP, climate warming could accelerate vegetation green-up via following mechanisms: First, climate warming induced the advance of SOS (Piao et al., 2006; Zhou et al., 2001). Earlier start of vegetation photosynthesis would enhance leaf C accumulation and thus form a positive feedback (Chapin, 1991), resulting in the increasing V_{NDVI} in early GUP. Second, vegetation growth is usually limited by energy in spring because of the low temperature (Chapin, 1991; Chapin et al., 2002), hence warming would alleviate the energy restriction, then increase vegetation productivity and accelerate vegetation greening (Zhu et al., 2016). Third, increasing temperature in spring could increase soil moisture and nutrient availability through enhancing the thawing of frozen soils in tundra ecosystems (Finger et al., 2016; Keuper et al., 2012), further accelerates vegetation greening.

However, these mechanisms cannot persist to the late GUP when vegetation growth is no longer limited by heat. On the contrary, climate warming can directly inhibit photosynthesis if the temperature exceeds the temperature optima of vegetation productivity, and consequently decelerated vegetation green-up

(Chen et al., 2021; Huang et al., 2019). In addition, the acceleration of vegetation green-up in early GUP may lead to premature depletion of soil moisture and nutrient (Lian et al., 2020; Zhang et al., 2021), and thus result in soil drought and nutrient deficiency, slowing down the velocity of vegetation green-up in late GUP. This effect would be further enhanced by increasing temperature, which increases atmospheric vapor pressure deficit and intensifies the water stress (Yuan et al., 2019). On this occasion, vegetation would allocate less carbon to organs consuming water (i.e., leaves) but more carbon to organs for water acquisition and transportation (e.g., roots and stems), which also results in the deceleration of vegetation green-up (Chapin et al., 2002). Indeed, the negative effects of warming on V_{NDVI} were observed earlier in the water-limited ecosystems of the inland areas, where soil moisture showed strong and positive effects on vegetation green-up (Figure 5). Interestingly, such divergent responses of vegetation green-up to climate warming were also found among different regions, especially for the temperature-constrained ecosystems vs. arid and semiarid ecosystems (Wang et al., 2018). Combining our study with the results in Wang et al. (2018), it is likely that climate warming accelerates vegetation green-up in the temperature limited regions and/or stages, but decelerates it where/when temperature is not major limiting factor.

Besides the shifting of limiting factors, the observed divergent changes of V_{NDVI} in different months may also reflect the movement of vegetation growth cycle (Park et al., 2015). Annual cycle of vegetation growth is impacted not only by environmental resource supplies but also by the longevity of leaf structure (Reich et al., 1992) and programmed cell-death in plants (Lim et al., 2007). Therefore, earlier leaf unfolding would also lead to the advance of programmed leaf life cycle and further result in different changes of V_{NDVI} in early and late GUP. As our analyses were conducted based on grid NDVI product, the mixed pixel effect would make the changes of V_{NDVI} more complicated (Chen et al., 2018). For instance, it is well known that increasing temperature can accelerate plant growth via temperature-dependent enzymatic catalytic reactions if temperature is lower than temperature optima (Atkinson & Porter, 1996). However, the optimal temperature of different species in a same pixel may be various. The different temperature optima and asynchronous movements of growth cycles among different species would generate a nonlinear response of V_{NDVI} to climate change at landscape or larger levels (Chen et al., 2018).

Across the mid-high-latitude Northern Hemisphere, areas with earlier starts of green-up usually showed slower green-up path (Figures 3 and 4). In other words, spatially, higher spring temperature leads to earlier start of green-up but negative effect on its velocity. This paradox was also found by previous studies (Clark et al., 2015; Seyednasrollah et al., 2018). Moreover, the temporal trends of V_{NDVI} and their relationships with climate change showed high spatial heterogeneity. Specifically, in areas with relatively shorter GUP, the overall temporal changes in V_{NDVI} were clear, that is, acceleration in the early stage but deceleration in the last stage (Figure 3). In contrast, taking the areas with longer GUP (>3 months) as a whole, V_{NDVI}

showed positive trends in the early months but then negative trends in following months, and positive trends in the last months during the GUP, which may be resulted from more complicated feedbacks between vegetation and environment (climate and soil). Therefore, only focusing on the trends in vegetation greenness is far from enough to understand the response of vegetation to climate change because many important details are masked.

The portfolio of change in the velocity of vegetation green-up (Figure 6) can better describe the complicated vegetation dynamics during GUP. Our results indicate that the early GUP (especially the SOS month) is a key stage that regulates the peak NDVI in summer. Vegetation green-up acceleration in SOS month generally results in a higher vegetation greenness in summer, which can be widely observed in Eurasia. In contrast, in a large part of North America, vegetation green-up deceleration in SOS month resulted in vegetation browning (negative NDVI trend) in past decades. Such deceleration was likely caused by the decreasing temperature in early spring (Figure S11). These results may result from the vegetation carryover effect (Lian et al., 2021) and demonstrated that vegetation green-up acceleration in the first month of GUP was an important contributor to the recently observed annual greening trend. Like an old saying in Chinese “The whole year’s work depends on a good start in spring”, our results highlight the importance of vegetation growth in early spring to peak vegetation productivity.

It is also important to highlight, however, that our results could involve some uncertainties. Given that our analyses were conducted at monthly level, the advance of leaf unfolding could also induce the lengthening of growth period in the first month, which also increases V_{NDVI} . Therefore, it is important to investigate green-up trajectories using data with higher temporal resolution in future studies. Nevertheless, the overall acceleration of green-up during the whole GUP (Wang et al., 2018) and the deceleration of green-up in the late stage again confirmed the robustness of green-up acceleration in early spring. It is also noteworthy that SOS derived from satellite data is generally much later than date of leaf unfolding (Gordo & Sanz, 2009), so the bias induced by changes in leaf unfolding date may be negligible.

In summary, our study revealed the divergent trends in the velocity of vegetation green-up in different months of GUP and their responses to climate change. These results suggest that the velocity of vegetation green-up can be more sensitive to climate change than vegetation greenness (e.g., NDVI, LAI) at monthly scale and thus can be used to understand the dynamics of vegetation growth at sub-seasonal or even finer scale. Moreover, our study also suggests that the first month of GUP is the key stage that determines vegetation productivity in summer and potentially in the whole year. Accordingly, green-up acceleration in the first month of GUP contributed largest to global greening trend over the past decades. Given the different magnitude of climate change and divergent responses of V_{NDVI} to climate change in different months of GUP, it is critically important to further explore the spatiotemporal heterogeneity in the velocity of vegetation green-up and its climate drivers through more long-term observation and more accurate modelling

approach for understanding and predicting the climate-vegetation feedbacks.

AUTHOR CONTRIBUTIONS

Songbai Hong and Yao Zhang designed the research; Songbai Hong performed the data analysis; Songbai Hong and Yao Zhang contributed to the interpretation of the results; Songbai Hong drafted the manuscript; all authors participated in discussions and the editing of the manuscript.

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CONFLICT OF INTEREST

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

All data used in this study are openly available from the following: GIMMS NDVI are available at <http://poles.tpdc.ac.cn/en/data/9775f2b4-7370-4e5e-a537-3482c9a83d88/>. CRU air temperature is available at https://crudata.uea.ac.uk/cru/data/hrg/cru_ts_4.05/cruts.2103051243.v4.05/; C3S soil moisture is available at <https://cds.climate.copernicus.eu/cdsapp#!/dataset/satellite-soil-moisture?tab=overview>; CRU-JRA v2.2 solar radiation is available at <https://catalogue.ceda.ac.uk/uuid/7f785c0e80aa4df2b39d068ce7351bbb>. MODIS EVI is available at <https://lpdaac.usgs.gov/products/mod13c2v006/>. All computer codes used in this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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